پـــاورلايـــن
POWERLINE



CERTIFIED ABB PANEL BUILDER · IEC 61439

# LV PANELS

Fully type-tested low voltage boards up to 6300 A and short circuit up to 120 kA

**PLP · PRO-E · Minicenter · UniKIT · SR Basic**

**Built by Powerline where it counts — partnered for the rest.**

- **01 PLP** Local modular MDB · 5000 A
- **02 PRO-E** MDB system · up to 6300 A
- **03 MINICENTER** Up to 24 modules
- **04 UNIKIT** Enclosures · up to 630 A
- **05 SR BASIC** Automation & distribution

| | | | |
|---|---|---|---|
| **6300 A** RATED CURRENT UP TO | **120 kA** SHORT CIRCUIT UP TO | **5000 A** EEHC-APPROVED OWN MDB | **ABB** CERTIFIED PANEL BUILDER |

01 · COMPANY PROFILE

# BUILT BY POWERLINE WHERE IT COUNTS — PARTNERED FOR THE REST

Powerline is a **certified ABB panel builder** offering fully type-tested solutions for low voltage boards up to **6300 A** and short circuit up to **120 kA**, in compliance with IEC 61439-1, IEC 61439-2 and IEC 61439-3 standards and ABB guidelines.

Since our inception in 2012, Powerline has grown from a trading company into a fully integrated engineering partner: an Original Equipment Manufacturer for medium voltage switchgear and a certified panel builder for low voltage boards, backed by two factories — the MV factory in 10th of Ramadan C3 and the LV & CSS factory in A5 — and more than 2,200 delivered projects.

### Two boards, one standard

Our own **PLP** main distribution board — type tested by Powerline and approved by the Egyptian Electricity Holding Company up to 5000 A — sits alongside the ABB **PRO-E** system up to 6300 A. Whichever the specification calls for, the engineering standard is the same.

### Local manufacturing, global partners

Boards are built and tested in Egypt with devices from ABB and Schneider Electric — short lead times and local support, without compromising on the type-test evidence behind the assembly.

## Quality is not a department — it is our work culture

### Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.

### Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.

### Testing & Quality Control

Every board leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

| 2012 | 2 | 2,200+ | ABB |
|---|---|---|---|
| ESTABLISHED · CAIRO | FACTORIES · MV AND LV/CSS | PROJECTS DELIVERED | CERTIFIED PANEL BUILDER |

" Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. "

LV PANELS · LOW VOLTAGE ASSEMBLY

# TABLE OF CONTENTS



**How to use this catalogue**

Sections 02–03 introduce the range and its certification. Sections 04–11 present each product with its data, standard sizes and components, in the order of our product line. Sections 12–15 cover application, selection, standards and construction.

02 · PRODUCT OVERVIEW

# THE LV PANEL RANGE

Five products cover the low voltage installation — two main distribution board systems and three distribution and enclosure ranges. All engineered, assembled and tested by Powerline.

| # | Product | Description | Rating | Role |
|---|---|---|---|---|
| 01 | PLP | Local modular type-tested low voltage distribution panel for power distribution and motor control — **Type Tested by Powerline** | EEHC approved to 5000 A | Main board |
| 02 | PRO-E | Modular type-tested main distribution board system compliant with IEC 61439-1 & 2 | up to 6300 A | Main board |
| 03 | Minicenter | Distribution boards for the residential and commercial segments, in different designs | up to 24 modules | Final / sub-main |
| 04 | UniKIT | Innovative enclosure system offering maximum functionality, simplicity, flexibility and quick assembly | 250 A wall / 630 A floor | Sub-main |
| 05 | SR Basic | Low voltage and medium-sized enclosures for automation and distribution, with a high degree of protection | IP65 · IK10 | Automation |


**PLP**


**PRO-E**


**Minicenter**


**UniKIT**


**SR Basic**

**6300 A**
BOARDS UP TO

**120 kA**
SHORT CIRCUIT UP TO

**61439**
IEC -1 / -2 / -3

**2**
MDB SYSTEMS TO CHOOSE FROM

**Main Distribution Board System PRO-E up to 6300 A — and Powerline's own MDB design, approved from EEHC up to 5000 A**

Two routes to the same result: the internationally established PRO-E platform, or PLP — our own locally manufactured design, type tested by Powerline and approved by the Egyptian Electricity Holding Company.

03 · CERTIFICATION

# TYPE-TESTED & CERTIFIED

A low voltage board is only as good as the evidence behind it. Every Powerline LV assembly is built on design-verified systems and routine-tested before it leaves the factory.

### Type Tested by Powerline

The PLP platform is designed and type tested by Powerline, verified against IEC standards for safety and performance — and approved by the Egyptian Electricity Holding Company up to 5000 A.

### Certified ABB panel builder

Powerline builds ABB systems to ABB guidelines, keeping the original type-test validity of the assembly — the certified combination of cabinet, busbars and devices.

### IEC 61439 compliance

Boards comply with IEC 61439-1 (general rules), -2 (power switchgear assemblies) and -3 (distribution boards for ordinary persons), as applicable to the product.

## What design verification proves

| Verified property | Why it matters on site |
|---|---|
| Temperature rise | The board carries its rated current continuously without overheating devices or cables |
| Short-circuit withstand | Busbars, supports and enclosure survive the prospective fault current — up to 120 kA |
| Dielectric properties | Clearances and insulation hold the rated impulse and power-frequency voltages |
| Protection against electric shock | Protective circuit continuity and effective bonding of every accessible metal part |
| Mechanical operation & IP | Doors, interlocks and covers work as designed and keep the declared IP rating |

**61439-1**
GENERAL RULES

**61439-2**
POWER ASSEMBLIES

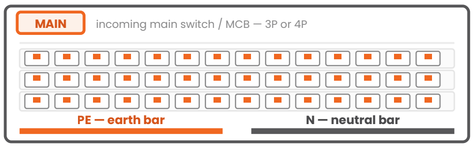
**61439-3**
DISTRIBUTION BOARDS

**EEHC**
APPROVED TO 5000 A

SECTIONS 05 – 06

# Main distribution boards

Two fully type-tested main distribution board systems for power distribution and motor control in commercial, industrial and infrastructure applications.




DIN-rail rows for outgoing MCBs — PE + N bars as standard

INCOMING DEVICE, BUSBAR SYSTEM, OUTGOING WAYS

**PLP** · pages 07–09
Local manufacturing · Type Tested by Powerline · EEHC approved to 5000 A

**PRO-E** · pages 10–12
Modular system up to 6300 A · IEC 61439-1 & 2

**6300 A**
PRO-E UP TO

**5000 A**
PLP · EEHC APPROVED

**120 kA**
SHORT CIRCUIT UP TO

**4b**
FORMS UP TO

05 · PLP

# PLP MAIN DISTRIBUTION BOARD

PLP is our local, fully type-tested low voltage distribution panel designed for power distribution and motor control in commercial, industrial and infrastructure applications. It is **Type Tested by Powerline**, verified against IEC standards for safety and performance.

| Attribute | PLP |
|---|---|
| Type | Modular, type-tested |
| Manufacturing | Local — Powerline LV factory, 10th of Ramadan A5 |
| Rated current | Up to 5000 A |
| Rated voltage | 400 V AC, 50 / 60 Hz |
| Material | Sheet metal, 2 mm thickness |
| Protection · impact | IP54 · IK10 |
| Standard | IEC 61439-1 |
| Approval | EEHC approved up to 5000 A |
| Partners | ABB & Schneider Electric devices |
| Application | Power distribution & motor control |

PLP main distribution board — Powerline LV factory

**5000 A**
EEHC APPROVED TO

**Local**
MANUFACTURING

### Highlights

- **Type Tested by Powerline** — our own design, our own verification
- Modular type-tested design — sections combine to any single-line diagram
- Local manufacturing — short lead times and direct factory support
- Suitable for commercial, industrial and infrastructure applications
- Available as customised solutions per client specifications

**Designed & type tested by Powerline**

The PLP platform carries our own type-test evidence — the assurance that the complete assembly, not just its components, has been verified.

05 · PLP

# PLP TECHNICAL DATA

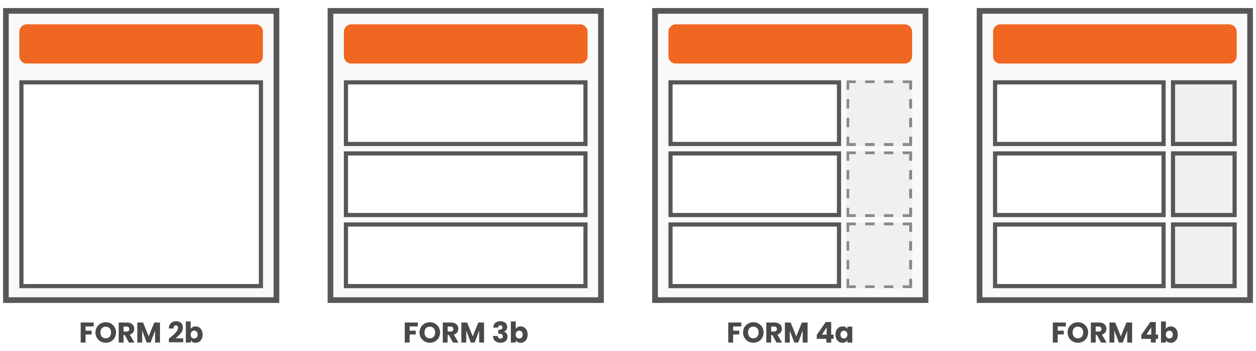
| Characteristic | PLP |
|---|---|
| Reference standards | IEC 61439-1 & -2 |
| Rated voltage | 400 V AC, 50 / 60 Hz |
| Rated current | up to 5000 A (EEHC approved) |
| Short-circuit withstand | per project, up to 120 kA platform limit |
| Material · thickness | Sheet metal, 2 mm |
| Degree of protection · impact | IP54 · IK10 |
| Mounting | Free standing |
| Incoming device | ACB or MCCB (ABB / Schneider) |
| Outgoing devices | MCCB, switch-fuse or motor starters |
| Internal separation | Form 2b / 3b / 4a / 4b |
| Busbars | Tinned copper, sized per rating |
| Enclosure | Sheet steel, RAL 7035 |
| Access | Front (front + rear for deep sections) |
| Applications | Distribution & motor control |

Guaranteed ratings, form of separation and busbar arrangement are confirmed per project at order stage.

| Standard sections — H × W × D (mm) |
|---|
| 2000 × 600 × 700 |
| 2000 × 600 × 900 |
| 2000 × 600 × 1100 |
| 2000 × 800 × 700 |
| 2000 × 800 × 900 |
| 2000 × 800 × 1100 |
| 2000 × 1000 × 700 |
| 2000 × 1000 × 900 |
| 2000 × 1000 × 1100 |

**Deeper sections** (900 / 1100 mm) take higher busbar ratings, rear cable access and full Form 4b compartmenting.

## Forms of internal separation




**Form 2b** — busbars separated from the functional units.

**Form 3b** — functional units separated from each other.

**Form 4a** — terminals in the same compartment as their unit.

**Form 4b** — terminals in their own separate compartment.

05 · PLP

# PLP TYPE TEST & APPROVALS

PLP is not an assembly of certified parts — it is a verified design. The platform has been type tested by Powerline against IEC requirements and approved by the Egyptian Electricity Holding Company for ratings up to 5000 A.

### Design verification — what was proven

| Verification | Result |
|---|---|
| Temperature-rise limits | Verified at rated current for the declared section arrangement |
| Short-circuit withstand strength | Busbar system, supports and enclosure verified at the declared Icw |
| Dielectric properties | Clearances, creepage and insulation verified for the rated voltages |
| Protective circuit continuity | Effective bonding of enclosure, doors and device frames |
| Mechanical operation & IP | Doors, covers and internal separation verified as built |

**Every board, every time**

On top of the design verification, each manufactured board passes the routine tests on page 23 and ships with its own test certificate and as-built documentation.

Type-test and approval certificates are issued with the project documentation; copies are available on request.

DESIGNED & TYPE TESTED BY

**POWERLINE**

Verified against IEC standards
for safety and performance

**EEHC approval**

Egyptian Electricity Holding Company approval for the PLP main distribution board design up to 5000 A.

**IEC**
VERIFIED DESIGN

**EEHC**
UTILITY APPROVAL

06 · PRO-E

# PRO-E MAIN DISTRIBUTION BOARD SYSTEM

PRO-E is a modular type-tested low voltage distribution panel designed for power distribution and motor control in commercial, industrial and infrastructure applications. It complies with IEC 61439-1 & 2, ensuring high safety, reliability and performance under various operating conditions.

| Attribute | PRO-E |
|---|---|
| Type | Modular, type-tested |
| Rated current | Up to 6300 A |
| Rated voltage | 400 V AC, 50 / 60 Hz |
| Protection · impact | IP65 · IK10 |
| Forms of segregation | 1, 2a, 2b, 3a, 3b, 4b |
| Standards | IEC 61439-1 & 2 |
| Application | Distribution & motor control |

PRO-E main distribution board — built by Powerline

**6300 A**
RATED CURRENT UP TO

**61439**
IEC -1 & -2

### Highlights

- Fully type-tested main distribution board system
- Compliant with IEC 61439-1 & 2 standards
- High safety, reliability and performance under various operating conditions
- Suitable for commercial, industrial and infrastructure applications
- Built by Powerline as a certified ABB panel builder, to ABB guidelines

**Two tiers, one platform**

Main distribution boards up to 6300 A, and the pro E energy wall-mounted and floor-standing sub-distribution cabinets up to 800 A on page 12 — a single system language across the installation.

06 · PRO-E

# PRO-E DATA & SECTIONS

| Characteristic | PRO-E main board |
|---|---|
| Reference standards | IEC 61439-1 & -2 |
| Rated current | up to 6300 A |
| Rated operational voltage | up to 415 V |
| Rated insulation voltage | 1000 V |
| Short-circuit withstand | up to 120 kA (platform) |
| Forms of segregation | 1 · 2a · 2b · 3a · 3b · 4b |
| Degree of protection · impact | IP65 · IK10 |
| Mounting · material | Free standing · sheet metal |
| Incoming device | ACB up to 6300 A |
| Outgoing devices | MCCB, switch-fuse, motor starters |
| Construction | Modular sheet-steel frame, RAL 7035 |
| Application | Distribution & motor control |

Ratings are platform maxima; the guaranteed values for a specific board depend on the section arrangement, device selection and ambient conditions, and are confirmed at order stage.

| Floor-standing sections — H × W × D (mm) |
|---|
| 2231 × 699 × 847 |
| 2231 × 699 × 1047 |
| 2231 × 899 × 847 |
| 2231 × 899 × 1047 |
| 2231 × 1099 × 847 |
| 2231 × 1099 × 1047 |

**Modular by design**

Sections bolt together into a suite with continuous busbars; widths and depths follow the device and cable requirements of each functional unit.

**Motor control.** Fixed and withdrawable starter arrangements, including soft starters and drives, integrate into the same frame system.

06 · PRO-E

# PRO-E SUB-DISTRIBUTION CABINETS

For sub-main distribution the same platform offers wall-mounted and floor-standing cabinets up to **800 A**, certified together with ABB low voltage devices to IEC 61439-2 / -3 and IEC 62208.

| Characteristic | Version L | Version M |
|---|---|---|
| Degree of protection | up to IP43 | up to IP55 |
| Delivery form | Flat kit | Monoblock |
| Protection class | I | |
| Rated current (In) | 800 A | |
| Rated operational voltage (Ue) | 415 V | |
| Rated insulation voltage (Ui) | 1000 V | |
| Short-time withstand (Icw) | 32 kA / 1 s | |
| Busbar system rated current | 800 A in IP43 | |
| Impact strength | IK07 · IK08 · IK09 per door type | |
| Cabinet widths / heights | 400 · 600 · 800 mm / 600 → 2000 mm | |
| Depth | 250 mm | |

### Materials & finish

| Part | Specification |
|---|---|
| Housing & door colour | RAL 7035 |
| Rear panel & profiles | Zinc-plated steel 1.5 mm |
| Side, top, bottom panels | Coated steel 1.25 mm |
| Plain / transparent door | Coated steel 1.25 mm / safety glass 3 mm |
| Cover plates | Coated steel 1 mm |
| Door opening angle | 130°, 3-point espagnolette locking |

### Panel-builder advantages

- Common busbars, functional units, doors, bases and flanges across both versions
- Quarter-turn screws pre-assembled; frame removes all covers at once
- Tool-free flange technology; doors ready to mount
- About 20 % more wiring space and easier accessibility

SECTIONS 08 – 10

# Distribution boards & enclosures

Boards and enclosures for the rest of the installation — from the apartment distribution board to the industrial automation enclosure.



SR BASIC ENCLOSURE FAMILY

| | | | |
|---|---|---|---|
| **24** | **630 A** | **IP65** | **IK10** |
| MINICENTER MODULES | UNIKIT UP TO | SR BASIC SINGLE DOOR | IMPACT RESISTANCE |

08 · MINICENTER

# MINICENTER DISTRIBUTION BOARDS

Minicenter — the successful future in residential and commercial segments, designed to provide safety and reliability. Available in different designs up to **24 modules**.

| Attribute | Minicenter |
|---|---|
| Segment | Residential & commercial |
| Capacity | Up to 24 modules |
| Focus | Safety & reliability |
| Reference standards | IEC 61439-1 & -3 |
| Maximum load current | 250 A |
| Maximum voltage | 240 / 415 V |
| Mounting | Flush (recessed in wall) |
| Incoming device | MCB / MCCB, 3P or 4P |
| Outgoing devices | Modular MCBs on busbar system |
| Degree of protection | IP41 · IK07 |
| Busbars | Included |
| Finish | RAL 9010 |


Minicenter distribution board

| Standard sizes — H × W × D (mm) |
|---|
| 540 × 420 × 132 |
| 600 × 420 × 132 |
| 650 × 420 × 132 |
| 760 × 420 × 132 |
| 870 × 420 × 132 |
| 1040 × 420 × 132 |
| 1180 × 420 × 132 |

**One opening, seven heights.** A common 420 mm width and 132 mm depth keeps the wall opening and the architectural detail identical across the range — only the module count changes.

09 · UNIKIT

# UNIKIT ENCLOSURE SYSTEM

UniKIT is an innovative enclosure system rated up to **630 A**, offering maximum functionality, simplicity, flexibility and quick assembly.

| Attribute | UniKIT |
|---|---|
| Type | Innovative enclosure system |
| Rated current — UniKIT 150, wall | Up to 250 A |
| Rated current — UniKIT 250, floor | Up to 630 A |
| Rated voltage | 400 V AC, 50 / 60 Hz |
| Reference standards | IEC 439-1 / EN 60439-1 / BS EN 60439-1 & 3 |
| Degree of protection | IP41 |
| Impact protection | IK10 |
| Finish | RAL 7035 |
| Maximum dimensions | 1900 × 800 × 300 mm |
| Mounting | Wall or floor standing |

UniKIT enclosure — wall-mounted version

**630 A**
RATED CURRENT

**IP54**
PROTECTION · IK10

## Highlights

- Maximum functionality — full internal layout flexibility for devices and busbars
- Simplicity and flexibility — one kit system covers many configurations
- Quick assembly — less time in the workshop, faster delivery to site
- Two variants: **UniKIT 150** wall-mounted up to 250 A and **UniKIT 250** free-standing up to 630 A

## Locally manufactured enclosure sizes — also available

| H × W × D (mm) · RAL 7035 · selection from 33 standard sizes | | | |
|---|---|---|---|
| 400 × 300 × 150 | 500 × 400 × 150 | 600 × 400 × 200 | 600 × 500 × 250 |
| 700 × 500 × 250 | 800 × 600 × 300 | 1000 × 600 × 300 | 1000 × 800 × 400 |
| 1200 × 800 × 300 | 1200 × 1000 × 300 | 1400 × 800 × 400 | 1400 × 1000 × 400 |
| 1600 × 800 × 400 | 1600 × 1000 × 400 | — | — |

Powerline also manufactures sheet-steel enclosures locally in 33 standard sizes from 400 × 300 × 150 mm to 1600 × 1000 × 400 mm — with custom sizes, cut-outs and short lead times on request.

10 · SR BASIC

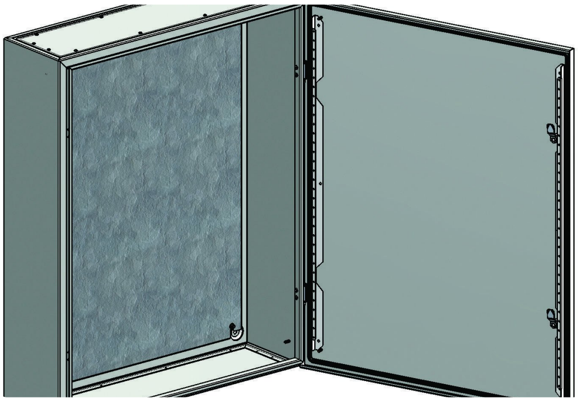
# SR BASIC ENCLOSURES

SR Basic offers low voltage and medium-sized enclosures for automation and distribution, with a high degree of protection in accordance with standard — a single line of metalwork that builds control boxes, secondary boards and end-distribution switchboards.

| Attribute | SR Basic |
|---|---|
| Type | LV & medium-sized enclosures |
| Use | Automation & distribution |
| Applications | Infrastructure, industrial, commercial |
| Rated voltage | 400 V AC, 50 / 60 Hz |
| Reference standard | IEC 62208 |
| Protection — single door | IP65 (IEC 60529) |
| Protection — double door | IP54 (IEC 60529) |
| Impact strength | IK10 (IEC 62262) |
| Dimensional range | H 300–2000 · W 200–1200 · D 150–400 mm |
| Material | 15/10 sheet steel · galvanised plate |
| Finish | RAL 7035 |

- Full apparatus range and modular breakers on DIN rails, or a blind back plate for automation
- Wall-mounted, floor-standing, stacked or coupled
- Continuous-cast polyurethane gasket · aluminium hinges · two door earth studs
- Removable gland plates top and bottom as standard

SR Basic enclosure built by Powerline



Wide mounting plate and continuous gasket

10 · SR BASIC

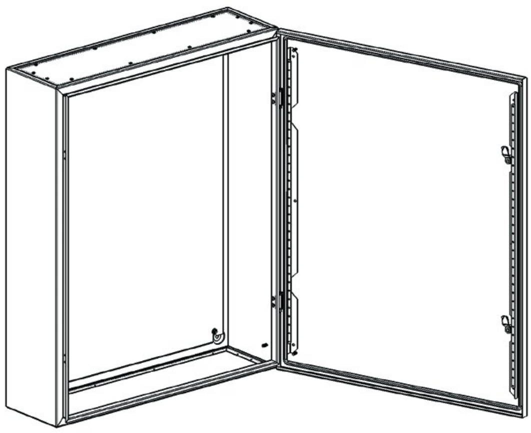
# SR BASIC STANDARD SIZES

All dimensions H × W × D in mm, finish RAL 7035. Single-door enclosures are rated IP65; double-door enclosures IP54. Supplied as empty enclosure with plain door, mounting plate and top and bottom gland plates.

| Single door — IP65 | |
|---|---|
| 300 × 200 × 150 | 300 × 300 × 150 |
| 400 × 300 × 150 | 400 × 300 × 200 |
| 500 × 400 × 150 | 500 × 400 × 200 |
| 600 × 400 × 150 | 600 × 400 × 200 |
| 600 × 400 × 250 | 600 × 600 × 250 |
| 700 × 500 × 200 | 700 × 500 × 250 |
| 800 × 600 × 200 | 800 × 600 × 250 |
| 800 × 600 × 300 | 800 × 800 × 300 |
| 1000 × 600 × 200 | 1000 × 600 × 250 |
| 1000 × 600 × 300 | 1000 × 800 × 200 |
| 1000 × 800 × 250 | 1000 × 800 × 300 |
| 1200 × 600 × 250 | 1200 × 600 × 300 |
| 1200 × 800 × 250 | 1200 × 800 × 300 |
| 1400 × 800 × 300 | 1400 × 800 × 400 |
| 1600 × 800 × 300 | 1600 × 800 × 400 |
| 1800 × 800 × 300 | 1800 × 800 × 400 |
| 2000 × 800 × 300 | 2000 × 800 × 400 |

| Double door — IP54 |
|---|
| 1000 × 1000 × 300 |
| 1200 × 1000 × 300 |
| 1200 × 1000 × 400 |
| 1400 × 1000 × 300 |
| 1400 × 1000 × 400 |
| 1600 × 1000 × 300 |
| 1600 × 1000 × 400 |
| 1800 × 1000 × 300 |
| 1800 × 1000 × 400 |
| 2000 × 1000 × 300 |
| 2000 × 1000 × 400 |
| 1800 × 1200 × 300 |
| 1800 × 1200 × 400 |
| 2000 × 1200 × 300 |
| 2000 × 1200 × 400 |

**49 standard sizes.** Enclosures may be wall-mounted, floor-standing or stacked; transparent doors, internal accessories, DIN-rail kits and alternative closing systems are available.



Enclosure with door open — outline

11 · COMPONENTS

# COMPONENTS & ACCESSORIES

Powerline is a certified ABB panel builder and works with **ABB and Schneider Electric** devices — with alternatives available where the consultant or client specifies them.

### Air circuit breakers (ACB)

Incoming protection up to 6300 A with electronic trip units, adjustable protection curves and communication options.

### Moulded case breakers (MCCB)

Incoming and outgoing protection for main and sub-main boards, thermal-magnetic or electronic trip.

### Modular MCBs & RCDs

Final circuit protection on DIN rails, with residual-current protection for socket and lighting circuits.

### Switch-fuse units

Fuse-protected outgoing ways where high breaking capacity and current limitation are required.

### Motor control

Contactors, overload relays, soft starters and drives for motor control centres in PLP and PRO-E boards.

### Metering & instruments

Ammeters, voltmeters, energy meters and multifunction power-quality meters with current transformers.

### Control & auxiliaries

Relays, timers, pilot devices, terminal blocks and internal wiring to project standard.

### Thermal management

Ventilation grilles, filtered fans, anti-condensation heaters and thermostats for warm or humid rooms.

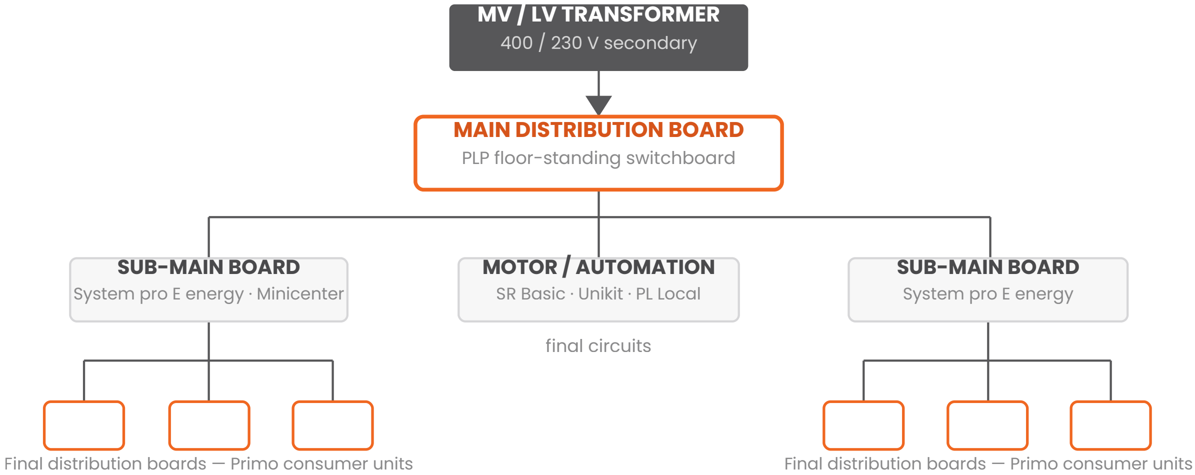
### Labelling & documentation

Engraved labels, mimic diagrams, circuit schedules, as-built drawings and test records with every board.

12 · APPLICATIONS

# FIELD OF APPLICATION

Low voltage distribution is a chain: the main board takes the transformer output, sub-main boards feed the zones and risers, and final boards protect the circuits people use. Powerline supplies every link.




Typical LV architecture — the Powerline product line at each level

## Typical duties

- Main distribution and motor control in commercial, industrial and infrastructure projects
- Sub-main distribution for floors, zones, risers and workshops
- Final distribution in residential and commercial buildings
- Automation, control and instrumentation enclosures

## Delivered across Egypt

- Malls & retail
- Hospitals
- Banks & offices
- Airports
- Water & utilities
- Factories
- Housing compounds
- Data centres
- Infrastructure

**A growing track record**

Hundreds of type-tested low voltage boards engineered, built and delivered for clients across Egypt — from single distribution boards to complete substation packages with our PCSS, PSEC and PRAL products.

13 · SELECTION

# HOW TO CHOOSE

Three questions decide the product: what role does the board play, how much current must it carry, and how harsh is the environment.

## 1 · By role and rating

| Role | Product | Selection note |
|---|---|---|
| Main board (MDB) | PLP | Locally manufactured, type tested by Powerline, EEHC approved up to 5000 A — power distribution and motor control |
| Main board (MDB) | PRO-E | Modular type-tested system up to 6300 A to IEC 61439-1 & 2 — where an international platform is specified |
| Sub-main board | UniKIT | Enclosure system up to 630 A with quick assembly and flexible internal layout |
| Final distribution | Minicenter | Residential and commercial boards, different designs up to 24 modules |
| Automation / control | SR Basic | Small and medium enclosures with a high degree of protection, wall or floor mounted |

## 2 · By protection degree

| IP rating | Product | Environment |
|---|---|---|
| IP41 | Minicenter | Indoor, clean, flush-mounted |
| IP43 | PRO-E sub-distribution L | Indoor sub-distribution |
| IP54 | UniKIT · SR Basic double door | Dust and splashing water |
| IP55 | PRO-E sub-distribution M | Indoor / sheltered outdoor |
| IP65 | SR Basic single door | Washdown, dusty, industrial |

## 3 · By form of internal separation

**Form 2b** — busbars separated from functional units. The economical choice for simple distribution duty.

**Form 3b** — units separated from each other. Work on one feeder without exposing the neighbours.

**Form 4a / 4b** — terminals separated too. Required where maintenance happens with the board live.

14 · STANDARDS

# STANDARDS & SERVICE CONDITIONS

Powerline LV boards comply with the IEC 61439 series and ABB guidelines; the enclosures follow IEC 62208, with IP and IK ratings verified to IEC 60529 and IEC 62262.

| Standard | Scope |
|---|---|
| IEC 61439-1 | Low voltage switchgear and controlgear assemblies — general rules |
| IEC 61439-2 | Power switchgear and controlgear assemblies — main and sub-main boards |
| IEC 61439-3 | Distribution boards intended to be operated by ordinary persons |
| IEC 62208 | Empty enclosures for low voltage switchgear and controlgear assemblies |
| IEC 60529 | Degrees of protection provided by enclosures (IP code) |
| IEC 62262 | Protection against external mechanical impacts (IK code) |
| ABB guidelines | Certified panel-builder rules preserving the validity of the original type tests |

## Service conditions

| Condition | Specification |
|---|---|
| Installation | Indoor |
| Ambient air temperature | -5 °C to +40 °C |
| Operating / storage maximum | +55 °C |
| Storage minimum | -40 °C |
| Relative humidity | Max. 95 % |
| Standard finish | RAL 7035 · RAL 9010 (Minicenter) |
| Factory acceptance test | Powerline standard FAT |

## Why it matters

- **Verified, not estimated** — temperature rise, short-circuit strength and dielectric properties are proven at assembly level.
- **Panel-builder discipline** — ABB systems built to ABB rules keep their type-test validity; deviations do not.
- **Local approvals** — the PLP platform carries EEHC approval, the reference for utility-facing projects in Egypt.

> Outside standard conditions — higher ambient, outdoor exposure, corrosive or dusty atmospheres — contact Powerline engineering for a derated or specially finished design.

15 · CONSTRUCTION

# CONSTRUCTION & QUALITY

### Sheet-steel bodies

- Epoxy-polyester powder coated sheet steel, RAL 7035
- Enclosure sheet 1.25 – 1.5 mm per family
- Zinc-plated rear panels and mounting profiles
- Galvanised mounting plates for apparatus
- Aluminium hinges on SR Basic for corrosion resistance

### Doors & access

- Plain or transparent doors; 130° opening angle
- Espagnolette closure with 3-point locking
- Swing handle or double-bit pin per mounting type
- Doors reversible left or right hinging
- Earth studs bond the door to the enclosure

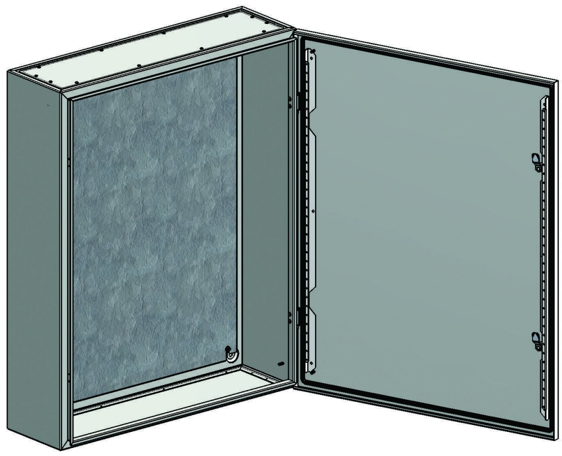
### Interior & wiring

- Tinned copper busbar systems sized to the rating
- DIN rails, mounting plates and quarter-turn cover plates
- Removable gland plates top and bottom
- PE and N bars on distribution boards
- Generous, accessible wiring space

### Sealing, impact and thermal design

Continuous-cast polyurethane gaskets give repeatable IP performance; knock-outs, flanges and gland plates preserve the rating after cable entry. Mechanical impact classes run from IK07 with knock-outs up to IK10 on the enclosure ranges. Where ambient temperatures or load losses demand it, ventilation grilles, filtered fans, heaters and thermostats are fitted.

**Labelling & documentation.** Engraved labels, mimic diagrams, circuit schedules, as-built drawings and test records are supplied with every board.



Enclosure interior — wide mounting plate, continuous gasket, removable gland plates

16 · QUALITY

# QUALITY, TESTING & ORDERING

## Routine verification — every board

- Inspection of the assembly, wiring and information (labels, diagrams, schedules)
- Dielectric test on main and auxiliary circuits
- Verification of protective-circuit continuity and earth bonding
- Functional test of devices, interlocks, indication and metering
- Verification of clearances, creepage and terminal torques
- Documented factory acceptance test — open to client witnessing

**Design verification behind every product**

PLP is type tested by Powerline and EEHC approved to 5000 A; PRO-E and the ABB enclosure ranges are type-tested systems built to ABB guidelines — verified at assembly level, not estimated.

## Information required when ordering

| # | Please specify |
|---|---|
| 1 | Product (PLP, PRO-E, Minicenter, UniKIT, SR Basic) and role |
| 2 | Rated current, voltage and prospective short-circuit current |
| 3 | Incoming device type and rating; outgoing ways with ratings |
| 4 | Form of internal separation (main and sub-main boards) |
| 5 | Mounting (flush, wall, floor), size limits and available space |
| 6 | Required IP / IK degree and ambient conditions |
| 7 | Device brand preference (ABB / Schneider Electric) |
| 8 | Metering, communication and accessory requirements |
| 9 | Single-line diagram and quantity |

**Note.** Ratings and dimensions in this catalogue are standard platform values subject to continuous product improvement and confirmed per order. Customised solutions are available per client specifications.

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# Built by Powerline where it counts — partnered for the rest.

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**6300 A**
BOARDS UP TO

**120 kA**
SHORT CIRCUIT UP TO

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-1 / -2 / -3

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